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Article

Pseudo-Hermitian Probability Across Admissible Projected Sectors

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Abstract

We study when different admissible projected realizations represent the same physical probability theory, and what obstructs their equivalence. Each member of a supplied family has a complex state sector, represented observable algebra, declared retained generator, and positive compatible metric. Strong operational equivalence intertwines these physical structures and preserves probabilities on equivalence classes of pointed preparation and measurement data. Necessary invariants include representation kernels, commutants, and the full spectral measures of physicalized generators. For a full matrix observable algebra, irreducible unital physical representations of the same abstract self-adjoint dynamical element yield a finite-dimensional sufficient equivalence criterion. Dimension-three examples exhibit distinct equivalent projections and an individually admissible sector excluded from their strong equivalence class by a spectral mismatch. Normalization, Gleason transport and matrix-algebra representation equivalence are standard ingredients used within this conditional framework. Projective probability retains explicit complex-dimension, additivity and preparation hypotheses. Neither projection classification nor metric uniqueness is established, and no probability measure on projection labels is supplied.

Keywords: pseudo-Hermitian quantum mechanics; quantum probability; projected sectors; operational equivalence; Gleason theorem; observable algebra

1. Introduction

The Born rule assigns probabilities to outcomes in a specified physical Hilbert space [2,3]. In pseudo-Hermitian quantum mechanics the auxiliary inner product need not be the physical one: a positive compatible metric can make retained evolution unitary [7,8]. Real spectra in PT-symmetric models do not, by themselves, specify that metric or its observable interpretation [4,11,12]. Our question is: when do different admissible projected realizations represent the same physical probability theory, and what obstructs such equivalence? Distinct state projectors need not encode different physical realizations, but individual admissibility alone does not establish their equivalence.

Projected models can distinguish enlarged kinematic data from probability-bearing state sectors. PT-symmetric models with quaternionic bookkeeping provide one motivation [1]; the mathematical results require only the supplied Hilbert, observable, generator and metric data below. No geometric construction or preferred projection is a theorem premise.

Genuine quaternionic Hilbert-space quantum mechanics has its own probability theory [9,10]. Moretti and Oppio formulate the quaternionic Gleason result with the appropriate trace structure [13]. In motivating models quaternionic structure may belong to enlarged kinematic data. Here probability is assigned on a supplied projected *complex* Hilbert realization. We prove no quaternionic Gleason theorem.

The pseudo-Hermitian conservation identity, the Gleason representation theorem, and probability invariance under physical isometries are individually standard. The present work isolates a different structural question: when is nonuniqueness in a family of admissible projected realizations merely representational, and when can it remain operationally substantive? We distinguish projection

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freedom from metric freedom, specify strong operational equivalence of complete realizations, and identify explicit obstructions showing that sector nonuniqueness need not reduce to representation freedom. Probability is well defined on equivalence classes of pointed admissible experimental data, including preparation and measurement. Alongside necessary obstructions, we give a sufficient finite-dimensional criterion using irreducible representations of a full matrix algebra and a common abstract dynamical element. These are structural uses of standard mathematics, not new general representation or Gleason theorems or a complete classification of projected realizations. A published operator example [1] also requires separating state-sector selection from the prescription that supplies retained dynamics. The paper's results do not determine which sector is realized and do not introduce a measure on the index set.

Section 2 specifies the admissibility package and separation of levels. Section 3 separates normalization from projective probability. Section 4 proves the comparison criterion, and Section 5 gives the worked example. Section 6 collects the scope boundaries.

2. Admissible Projected Realizations

2.1. State Sectors and Represented Observables

For the abstract construction fix a complex auxiliary Hilbert representation $\mathcal{H}_{\mathrm{kin}}$ of the enlarged carrier. Such an auxiliary representation does not automatically carry physical probability. Consider a declared family of bounded, time-independent idempotents

$$\mathcal{P}_{\mathrm{adm}} = \{\Pi_\alpha\}_{\alpha\in\mathcal{I}}, \qquad \Pi_\alpha^2 = \Pi_\alpha, \qquad \mathcal{H}_\alpha = \operatorname{Ran}\Pi_\alpha. \tag{1}$$

Their ranges are closed. Orthogonality in the auxiliary pairing is not required. The notation $\mathcal{P}_{\mathrm{adm}}$ is reserved for candidates supplied with the package below; it does not assert that the family is exhaustive, has multiple members in a particular motivating model, or is unique. No measure or measurable structure on $\mathcal{I}$ is assumed.

Supply a unital abstract observable $*$-algebra $\mathcal{A}_{\mathrm{obs}}$ and bounded sector representations

$$\pi_\alpha : \mathcal{A}_{\mathrm{obs}} \longrightarrow \mathcal{B}(\mathcal{H}_\alpha), \qquad \mathcal{A}_\alpha = \pi_\alpha(\mathcal{A}_{\mathrm{obs}}). \tag{2}$$

Faithfulness of π_α is not assumed unless required in a particular comparison. This explicitly distinguishes state selection from observable representation. When a comparison is made, the same abstract algebra specifies which operations are compared. It need not be the full algebra of all bounded operators. Self-adjoint abstract observables are represented as self-adjoint operators in the chosen physical inner product. The available observable algebra and the all-projector measure hypothesis in the later Gleason corollary are separate assumptions.

Definition 1 (Admissibility package)**.** *A probability-admissible projected realization supplies the following data and hypotheses:*

1. *a compatible complex Hilbert realization on $\mathcal{H}_\alpha$, whose represented observables preserve that sector;*
2. *a declared retention/reduction prescription giving a densely defined, time-independent generator*

$$H_{\mathrm{ret},\alpha} : \mathcal{D}_\alpha \subseteq \mathcal{H}_\alpha \longrightarrow \mathcal{H}_\alpha; \tag{3}$$

   selection is not identified with exact elimination of a coupled complement;
3. *a bounded, positive, boundedly invertible, time-independent metric G_α, with physical inner product and adjoint*

$$\langle x \mid y\rangle_{G_\alpha} = \langle x \mid G_\alpha y\rangle, \qquad B^{\sharp_\alpha} = G_\alpha^{-1} B^\dagger G_\alpha, \tag{4}$$

   such that $\pi_\alpha(O^*) = \pi_\alpha(O)^{\sharp_\alpha}$ and

$$H_{\mathrm{ret},\alpha}^\dagger G_\alpha = G_\alpha H_{\mathrm{ret},\alpha}; \tag{5}$$

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4. *self-adjointness of $H_{\mathrm{ret},\alpha}$ in the G_α-Hilbert space. For unbounded generators Eq. (5) includes equality of operator domains, equivalently $\mathcal{D}(H_{\mathrm{ret},\alpha}^\dagger) = G_\alpha \mathcal{D}_\alpha$. Evolution is the corresponding unitary group; solutions starting in $\mathcal{D}_\alpha$ remain there and are differentiable.*

Projective probability additionally requires the dimension, measure, and preparation hypotheses of Corollary 1. These are not consequences of the geometric observable filter or of norm conservation.

This is a sufficient package for the results studied here, not a classification of all physically possible projections. Self-adjointness is stronger than a formal differential operator identity; no general existence or domain theorem for a proposed effective generator is supplied. Units below are $\hbar = 1$.

2.2. Projection, Sector Preservation, and Effective Reduction

For an original generator whose evolution preserves $\mathcal{H}_\alpha$, strict restriction is an allowed retained prescription, with its proper restricted domain. A decoupled block operator is sufficient. The sandwich $\Pi_\alpha H_{\mathrm{aux}} \Pi_\alpha$ alone does not establish invariance of the original evolution or exact elimination. The published scalar differential-operator example [1] distinguishes

$$D_{\mathrm{strict}} = D_{++}, \qquad D_{\mathrm{eff},+} = D_{++} - D_{+-} D_{--}^{-1} D_{-+}. \tag{6}$$

The inverse and compositions require compatible domains and a common boundary/state prescription. That example has $D_{+-} = +B$, $D_{-+} = -B$, hence the correction is $+BD_{--}^{-1}B$. A complement can affect retained dynamics even though probabilities are assigned only within the declared sector. Strict and exact operators agree when the correction vanishes; neglecting it otherwise is a declared approximation. This differential-operator formula is not generalized here into a universal Hamiltonian theorem. Nonlocal or energy-dependent elimination need not yield an autonomous self-adjoint generator satisfying Definition 1.

2.3. Motivation and Separation of Levels

Projected PT-symmetric models with enlarged quaternionic data motivate separating kinematic bookkeeping from probability-bearing state sectors [1]. Scalar or geometric filtering does not construct the complex state projector or its observable representation. The selector Π_α and representation π_α are supplied separately through Definition 1. No particular trace-plus-compensator chart is required by the abstract mathematical results, and no unique geometric construction of Π_α is claimed.

2.4. Metric Freedom Within a Sector

Metric nonuniqueness is already standard quasi-/pseudo-Hermitian theory [8,11,12]. In finite dimension a diagonalizable real-spectrum generator with a complete biorthogonal basis admits

$$G_{\alpha,w} = \sum_n w_n |\chi_n\rangle\langle\chi_n|, \qquad w_n > 0, \quad \langle\chi_m \mid \phi_n\rangle = \delta_{mn}. \tag{7}$$

For $x = \sum_n c_n \phi_n$, its quadratic form is $\sum_n w_n |c_n|^2 > 0$ for $x \neq 0$, and the real eigenvalues give $H^\dagger G_{\alpha,w} = G_{\alpha,w} H$. This guarantees generator compatibility, not compatibility with every preassigned observable representation. Additional observables can constrain the weights.

The familiar irreducible-observable mechanism illustrates that further input [11,12]. If two bounded positive invertible metrics G, G' make the same bounded represented self-adjoint observables physical, then $R = G^{-1/2} G' G^{-1/2}$ commutes with their ordinary self-adjoint representatives $G^{1/2}\pi(O)G^{-1/2}$. A scalar commutant implies $R = cI$, hence $G' = cG$, $c > 0$. This is a standard conditional statement, not a metric-fixing theorem for motivating models; retained dynamics alone need not give that commutant condition.

Multiple admissible projections and multiple positive compatible metrics within a fixed sector are different freedoms. Distinct compatible metrics are not assumed physically equivalent. Equality of their sesquilinear forms on all states would just imply equality of the bounded metric operators,

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and supplies no useful broader notion of equivalence. Instead, Section 4 compares full realizations by explicit state, metric, observable, and dynamical transport.

3. Sectorwise Normalization and Projective Probability

Proposition 1 (Sectorwise pseudo-Hermitian normalization)**.** *For every α in the supplied admissible family, let $H_{\mathrm{ret},\alpha}, G_\alpha$ satisfy Definition 1. For $i\partial_t\psi_\alpha = H_{\mathrm{ret},\alpha}\psi_\alpha$ with initial state in $\mathcal{D}_\alpha$,*

$$\frac{d}{dt}\|\psi_\alpha(t)\|^2_{G_\alpha} = 0. \tag{8}$$

The unitary group preserves the norm also for arbitrary initial states in the completed physical Hilbert space. This is a conserved positive quadratic normalization functional; it is not yet a projective probability measure.

Proof. For each fixed α, differentiability, fixed G_α and compatible domains give

$$\frac{d}{dt}\langle\psi_\alpha \mid G_\alpha\psi_\alpha\rangle = i\langle H_{\mathrm{ret},\alpha}\psi_\alpha \mid G_\alpha\psi_\alpha\rangle - i\langle\psi_\alpha \mid G_\alpha H_{\mathrm{ret},\alpha}\psi_\alpha\rangle = 0.$$

The adjoint identity makes the terms equal. Self-adjointness in the physical inner product gives the unitary-group statement for all states. □

This is the standard pseudo-Hermitian identity applied independently to every supplied admissible sector. Its family indexing neither constructs the family nor imposes a relation between probabilities in different sectors.

Corollary 1 (Sectorwise Gleason measure and calibrated pure preparation)**.** *For each α independently suppose $\mathcal{H}_\alpha$ is separable and complex, $\dim_{\mathbb{C}}\mathcal{H}_\alpha \geq 3$. Let μ_α be a nonnegative normalized, noncontextual measure on* all *G_α-orthogonal projectors, with $\mu_\alpha(I_\alpha) = 1$ and countable additivity on mutually orthogonal projectors (their sum converges strongly). In finite dimension finite additivity suffices. Noncontextuality means a projector's value does not depend on the complete orthogonal resolution containing it. Then a unique positive, physical-self-adjoint trace-class density operator ρ_α of physical trace one represents the specified measure:*

$$\mu_\alpha(P) = \mathrm{Tr}_{G_\alpha}(\rho_\alpha P). \tag{9}$$

For a normalized pure prepared ray ψ_α, additionally impose certainty $\mu_\alpha(P_{\psi_\alpha}) = 1$, where $P_{\psi_\alpha}x = \psi_\alpha\langle\psi_\alpha \mid x\rangle_{G_\alpha}$. For a complete G_α-orthonormal family $\{\phi_i^{(\alpha)}\}$ this gives

$$P_i^{(\alpha)} = \mu_\alpha(P_i) = |\langle\phi_i^{(\alpha)} \mid \psi_\alpha\rangle_{G_\alpha}|^2, \qquad \sum_i P_i^{(\alpha)} = 1. \tag{10}$$

Here P_i on the measure's argument denotes the rank-one projector onto $\phi_i^{(\alpha)}$, not the numerical probability $P_i^{(\alpha)}$. This defines probability conditional on α. It does not define probability of α.

Proof. The positive square root $J_\alpha = G_\alpha^{1/2}$ is an onto isometry from the physical Hilbert space to its ordinary complex Hilbert realization. Every physical orthogonal projector maps to the ordinary orthogonal projector $\widehat{P} = J_\alpha P J_\alpha^{-1}$. The transported measure has the stated additivity, normalization and dimension hypotheses, so the standard Gleason theorem [3] gives $\widehat{\mu}_\alpha(\widehat{P}) = \mathrm{Tr}(\widehat{\rho}_\alpha\widehat{P})$. Set $\rho_\alpha = J_\alpha^{-1}\widehat{\rho}_\alpha J_\alpha$ and take the trace in the physical Hilbert space to obtain Eq. (9). The preparation certainty condition and positivity of a trace-one density imply $\widehat{\rho}_\alpha = |J_\alpha\psi_\alpha\rangle\langle J_\alpha\psi_\alpha|$: its positive square root annihilates the orthogonal complement of that ray. Thus $\rho_\alpha = P_{\psi_\alpha}$, and the rank-one projector expectation is the squared physical overlap in Eq. (10). □

The result transports ordinary Gleason theory; it is not a new Gleason theorem. The measure assumptions are probability input, and calibration is additional preparation input. A conserved norm

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alone supplies neither. If only a restricted collection of projectors is assigned measures, this corollary cannot be invoked without its all-projector hypothesis. Dimension two is excluded.

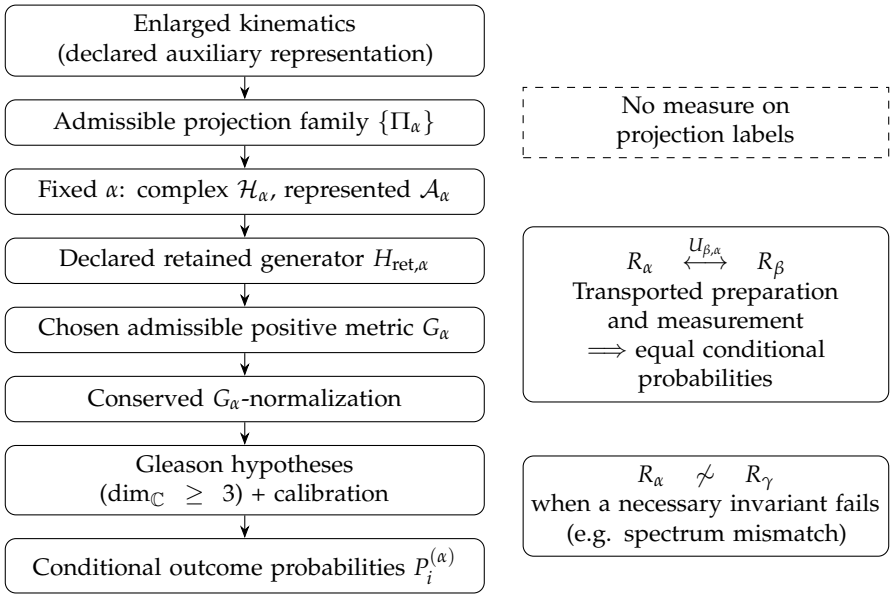


Figure 1. Data and hypotheses needed for a fixed-sector probability construction. Arrows denote supplied structures or conditional steps, not automatic geometric implications. Horizontal comparison concerns complete realization and experimental data; an obstruction excludes the stated strong equivalence.

4. Inter-Sector Operational Equivalence

Write a complete admissible projected realization as

$$R_\alpha = (\Pi_\alpha, \mathcal{H}_\alpha, \pi_\alpha, H_{\mathrm{ret},\alpha}, G_\alpha), \qquad \mathcal{R}_{\mathrm{adm}} = \{R_\alpha\}_{\alpha\in\mathcal{I}}. \tag{11}$$

The generator includes its declared prescription and domain. Fix the compared abstract observable algebra throughout a comparison. The relation below compares these complete physical data, not bare projectors or ambient embeddings.

Definition 2 (Strong equivalence relative to compared operations)**.** *Write $R_\alpha \sim R_\beta$ when the already admissible realizations are strongly operationally equivalent relative to the specified common bounded observable algebra: a bounded invertible map $U = U_{\beta,\alpha} : \mathcal{H}_\alpha \to \mathcal{H}_\beta$ satisfies*

$$U^\dagger G_\beta U = G_\alpha, \tag{12}$$

$$U\mathcal{D}_\alpha = \mathcal{D}_\beta, \qquad UH_{\mathrm{ret},\alpha}x = H_{\mathrm{ret},\beta}Ux \quad (x \in \mathcal{D}_\alpha), \tag{13}$$

$$U\pi_\alpha(O)U^{-1} = \pi_\beta(O) \quad (O \in \mathcal{A}_{\mathrm{obs}}). \tag{14}$$

Preparations and measurement projectors being compared must also be transported:

$$\rho_\beta = U\rho_\alpha U^{-1}, \qquad P_\beta = UP_\alpha U^{-1}. \tag{15}$$

This is a sufficient strong criterion, not a claim to characterize every possible weaker operational equivalence. Metric intertwining makes U unitary between the physical Hilbert spaces. It ensures that transported projectors remain physical orthogonal projectors and densities remain positive with trace one. For measurements belonging to the compared observable algebra, measurement transport follows from observable intertwining; other projectors are compared only when explicitly transported. If unbounded observables are to be compared, their domains must also be mapped; they are not silently included in Eq. (14).

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Proposition 2 (Probability invariance under physical intertwiners)**.** *For realizations satisfying Definition 2, take positive physical trace-class densities of trace one and physical orthogonal measurement projectors transported as in Eq. (15). Their probabilities obey*

$$\mu_\beta(P_\beta) = \mu_\alpha(P_\alpha). \tag{16}$$

For calibrated pure preparations $\psi_\beta = U\psi_\alpha$ and transported normalized measurement vectors $\phi_\beta = U\phi_\alpha$,

$$|\langle \phi_\beta \mid \psi_\beta \rangle_{G_\beta}|^2 = |\langle \phi_\alpha \mid \psi_\alpha \rangle_{G_\alpha}|^2. \tag{17}$$

These equalities persist under the intertwined retained evolutions.

Proof. Set $W = G_\beta^{1/2} U G_\alpha^{-1/2}$. Metric intertwining and invertibility give $W^\dagger W = WW^\dagger = I$. In ordinary Hilbert realizations, $\widehat{\rho}_\beta = W\widehat{\rho}_\alpha W^{-1}$ and $\widehat{P}_\beta = W\widehat{P}_\alpha W^{-1}$. Unitary invariance of the trace of a trace-class density times a bounded projector proves Eq. (16). Directly, $\langle U\phi_\alpha \mid U\psi_\alpha \rangle_{G_\beta} = \langle \phi_\alpha \mid U^\dagger G_\beta U \psi_\alpha \rangle = \langle \phi_\alpha \mid \psi_\alpha \rangle_{G_\alpha}$. Exact domain and generator intertwining for self-adjoint physical generators gives $Ue^{-itH_{\mathrm{ret},\alpha}} = e^{-itH_{\mathrm{ret},\beta}} U$, proving the evolution statement. □

This is the standard isometric-equivalence fact, used to organize admissible projected realizations. Identity, inverse, and composition preserve the defining conditions, so it provides an equivalence relation on $\mathcal{R}_{\mathrm{adm}}$ with a fixed compared algebra. Write $\mathcal{M}_{\mathrm{adm}} = \mathcal{R}_{\mathrm{adm}}/\sim$ for its operational equivalence classes. This quotient is organizational notation, not a classification result or an equivalence relation on bare projectors. The family-independent statement is the same conditional normalization and measure *form* in every sector. Numerical outcome probabilities are invariant under the specified transport; no numerical invariance for arbitrary inequivalent sectors or unrelated preparations is asserted.

4.1. Pointed Experimental Data and Equivalence Obstructions

An experimental datum is a pointed realization

$$X_\alpha = (R_\alpha, \rho_\alpha, P_\alpha), \tag{18}$$

where ρ_α is an admissible positive physical-self-adjoint trace-class density of physical trace one, and P_α is a physical orthogonal measurement projector. Declare $X_\alpha \sim X_\beta$ if a strong physical equivalence also transports both entries as in Eq. (15). Identity, inverse and composition preserve this relation. The admissible experimental data under comparison include these transported preparations and measurements.

Corollary 2 (Probability on equivalence classes of pointed experimental data)**.** *For the pointed data above,*

$$\overline{\mu}([X_\alpha]) = \mathrm{Tr}_{G_\alpha}(\rho_\alpha P_\alpha) \tag{19}$$

is independent of the representative. For calibrated pure preparations and rank-one measurements it equals $|\langle \phi_\alpha \mid \psi_\alpha \rangle_{G_\alpha}|^2$.

Proof. Any two representatives are related by the transport in Proposition 2, which preserves the trace and pure overlap. □

This direct consequence assigns an outcome probability to an equivalence class of pointed admissible experimental data, including preparation and measurement. It assigns no probability to $[R_\alpha] \in \mathcal{M}_{\mathrm{adm}}$ or to a projection label.

Proposition 3 (Necessary invariants of strong equivalence)**.** *Let $R_\alpha \sim R_\beta$, $J_\nu = G_\nu^{1/2}$, $\widetilde{\pi}_\nu(O) = J_\nu \pi_\nu(O) J_\nu^{-1}$ and $h_\nu = J_\nu H_{\mathrm{ret},\nu} J_\nu^{-1}$, $\nu = \alpha, \beta$. Hilbert dimensions agree and* $\ker \pi_\alpha = \ker \pi_\beta$. *The unitary*

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$W = J_\beta U J_\alpha^{-1}$ intertwines the physicalized represented algebras and transports their commutants. It intertwines the self-adjoint generators on their exact domains $J_\nu \mathcal{D}_\nu$. For their spectral projection-valued measures and every Borel set $\Delta \subseteq \mathbb{R}$,

$$E_\beta(\Delta) = W E_\alpha(\Delta) W^{-1}, \qquad f(h_\beta) = W f(h_\alpha) W^{-1} \tag{20}$$

for every bounded Borel function f in the second identity. Consequently spectra, eigenvalue multiplicities and spectral multiplicities wherever defined agree. Transported transition probabilities also agree.

Proof. The metric identity makes W onto unitary. Observable intertwining gives equal kernels and $W\widetilde{\pi}_\alpha(O)W^{-1} = \widetilde{\pi}_\beta(O)$, so conjugation transports the represented algebras and their commutants. Exact generator-domain intertwining gives $W h_\alpha W^{-1} = h_\beta$. Uniqueness of the self-adjoint spectral resolution yields the first identity in Eq. (20); integration of bounded Borel functions yields the second. These transports preserve the stated dimension and spectral data. Proposition 2 preserves physical overlaps. □

These are necessary invariants, not a complete classification. A spectral mismatch rules out strong equivalence. The criterion compares generators at a fixed time scale without an energy-origin shift. Its failure does not rule out every weaker equivalence based on a restricted set of experiments, nor does it canonically identify preparations or measurements between inequivalent realizations.

Proposition 4 (Finite-dimensional sufficient equivalence criterion)**.** *Let $\mathcal{A}_{\mathrm{obs}} = M_n(\mathbb{C})$ and let R_α, R_β be nonzero finite-dimensional admissible realizations. Suppose their physicalized representations $\widetilde{\pi}_\nu(O) = G_\nu^{1/2}\pi_\nu(O)G_\nu^{-1/2}$ are unital irreducible $*$-representations and, for the same $H_0 = H_0^* \in M_n(\mathbb{C})$,*

$$H_{\mathrm{ret},\nu} = \pi_\nu(H_0), \qquad \nu = \alpha, \beta. \tag{21}$$

Then $R_\alpha \sim R_\beta$. In particular, there is an ordinary unitary W intertwining the physicalized representations, and

$$U = G_\beta^{-1/2} W G_\alpha^{1/2} \tag{22}$$

satisfies Definition 2.

Proof. The standard irreducible representation theory of $M_n(\mathbb{C})$ gives unitary equivalence [14]. Since $M_n(\mathbb{C})$ is simple, every nonzero unital representation has trivial kernel and is faithful. Hence $\widetilde{\pi}(E_{11}) \neq 0$. Choose a unit vector v in its range. The vectors $\widetilde{\pi}(E_{i1})v$ are orthonormal and their span is invariant under all matrix units. Irreducibility makes this span the whole space, hence its dimension is n. Corresponding bases in the two representations define W with $W\widetilde{\pi}_\alpha(O)W^{-1} = \widetilde{\pi}_\beta(O)$ for every O. Writing $J_\nu = G_\nu^{1/2}$, substitution of $U = J_\beta^{-1} W J_\alpha$ gives

$$U^\dagger G_\beta U = J_\alpha W^\dagger W J_\alpha = G_\alpha, \qquad U\pi_\alpha(O)U^{-1} = \pi_\beta(O).$$

Taking $O = H_0$ yields $U H_{\mathrm{ret},\alpha} = H_{\mathrm{ret},\beta} U$. All generator domains are full finite-dimensional spaces, so domain mapping is automatic. These are all the strong-equivalence conditions. □

The representation-theoretic input is standard; its role here is a sufficient criterion for complete admissible projected realizations. It does not establish projection or metric uniqueness. Irreducible observable representations alone are insufficient without the common dynamical element. The criterion applies for $n \geq 1$; the separate Gleason probability hypotheses still require $n \geq 3$.

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5. Worked Finite-Dimensional Realizations

5.1. Dimension Three: the Full Conditional Construction

This synthetic construction illustrates the complete normalization/measure chain and the equivalence criterion; it is neither empirical evidence for a motivating model nor a geometric derivation of its state projections. Let

$$\mathcal{H}_{\mathrm{kin}} = \mathbb{C}^3 \oplus \mathbb{C}^3, \qquad \Pi_a = \begin{pmatrix} I_3 & 0 \\ 0 & 0 \end{pmatrix}, \quad \Pi_b = \begin{pmatrix} 0 & 0 \\ 0 & I_3 \end{pmatrix}. \tag{23}$$

For either selected sector the other summand is complementary bookkeeping; no probability measure on their labels is used. In the first sector choose

$$\begin{gathered} h = \mathrm{diag}(1,2,3), \quad S = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad S^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \\ H_a := H_{\mathrm{ret},a} = S^{-1}hS = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{pmatrix}, \qquad G_a = S^{\dagger}S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}. \end{gathered} \tag{24}$$

Declare strict restriction of the decoupled auxiliary generator $K_{\mathrm{kin}} = \mathrm{diag}(H_a, h)$. Thus both ranges are invariant, with $H_{\mathrm{ret},b} = h$, $G_b = I_3$ in the second sector, and all domains are the complete finite-dimensional spaces. Both spectra are $\{1,2,3\}$. Since $\det S = 1$, $x^{\dagger}G_a x = \|Sx\|^2 > 0$ for $x \neq 0$. The leading principal minors of G_a are $1,1,1$; explicitly,

$$H_a^{\dagger}G_a = G_a H_a = S^{\dagger}hS = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 2 & 5 \end{pmatrix}. \tag{25}$$

This is an explicit $n = 3$ illustration of Proposition 4. Supply $\mathcal{A}_{\mathrm{obs}} = M_3(\mathbb{C})$ with $\pi_a(O) = S^{-1}OS$, $\pi_b(O) = O$, and common $H_0 = h$. The physical adjoint respects this algebra's ordinary $*$; Hermitian O represent physical observables in both sectors. For $J_a = G_a^{1/2}$, the unitary $W = SJ_a^{-1}$ conjugates the physicalized representation in a to the defining irreducible representation in b. Thus both physicalized representations are irreducible, both retained generators represent $H_0 = h$, and Eq. (22) gives $U = S$. The columns $\phi_i^{(a)} = S^{-1}e_i$ are an eigenbasis satisfying $\langle \phi_i^{(a)} \mid \phi_j^{(a)} \rangle_{G_a} = \delta_{ij}$. Choose a calibrated pure preparation

$$u = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{3} \\ 1/\sqrt{6} \end{pmatrix}, \qquad \psi_a = S^{-1}u = \begin{pmatrix} 1/\sqrt{2} - 1/\sqrt{3} + 1/\sqrt{6} \\ 1/\sqrt{3} - 1/\sqrt{6} \\ 1/\sqrt{6} \end{pmatrix}. \tag{26}$$

Then $\|\psi_a\|_{G_a}^2 = \|u\|^2 = 1$, and $\psi_a(t) = S^{-1}e^{-ith}u$ has the same norm for all t by Proposition 1.

To instantiate Corollary 1, explicitly supply the nonnegative normalized additive measure on all physical orthogonal projectors with pure density $\rho_a = P_{\psi_a}$. It has preparation certainty and all dimension-three Gleason hypotheses. For the measured eigenprojectors $P_{i,a} = S^{-1}|e_i\rangle\langle e_i|S$,

$$\langle \phi_i^{(a)} \mid \psi_a \rangle_{G_a} = u_i, \qquad (P_1^{(a)}, P_2^{(a)}, P_3^{(a)}) = (\tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{6}), \qquad \sum_i P_i^{(a)} = 1. \tag{27}$$

This verifies the measure representation and its calibrated pure-state specialization; it does not manufacture the measure assumptions from conserved normalization.

In sector b take $\psi_b = u$, $\phi_i^{(b)} = e_i$. The local-coordinate inter-sector map $U_{b,a} = S$ obeys $U^{\dagger}G_b U = G_a$, $UH_a = hU$ and $U\pi_a(O)U^{-1} = \pi_b(O)$. It transports preparations and all projectors, giving the same three probabilities by Proposition 2. Here S is an explicit physical isometry, distinct from the

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positive-square-root map used in the general Gleason proof. Two distinct admissible projections in this synthetic model are equivalent under the specified operations; no statement about all projections in a motivating model follows.

5.2. An Individually Admissible, Inequivalent Third Sector

Preserve all the local a, b data and extend their carrier by a decoupled third summand:

$$\begin{gathered}\mathcal{H}_{\mathrm{kin}}^{(3)} = \mathbb{C}^3 \oplus \mathbb{C}^3 \oplus \mathbb{C}^3, \quad K_{\mathrm{kin}}^{(3)} = \mathrm{diag}(H_a, h, H_c), \\ \Pi_a^{(3)} = \Pi_a \oplus 0_3, \quad \Pi_b^{(3)} = \Pi_b \oplus 0_3, \quad \Pi_c^{(3)} = \mathrm{diag}(0_3, 0_3, I_3), \\ H_c := H_{\mathrm{ret},c} = \mathrm{diag}(1,2,4), \quad G_c = I_3, \quad \pi_c(O) = O \quad (O \in M_3(\mathbb{C})).\end{gathered} \tag{28}$$

Write R_a, R_b also for their zero-extended realizations. Strict restriction of this decoupled carrier supplies all three retained generators. Sector c has complex state space $\mathbb{C}^3$, full domain, bounded observable representation, positive metric and $H_c^\dagger G_c = G_c H_c$. Its norm is conserved. A positive trace-one density defines an additive measure on all orthogonal projectors; an independently calibrated normalized pure ray satisfies Corollary 1. Thus c is individually admissible, including the available dimension-three probability construction.

However,

$$\mathrm{spec}(H_a) = \{1,2,3\} \neq \{1,2,4\} = \mathrm{spec}(H_c), \qquad R_a \not\sim R_c. \tag{29}$$

Although its observable representation is irreducible, $H_c = \pi_c(H_c)$ does not represent the common element $H_0 = h$ used for a, b; Proposition 4 therefore does not apply to that comparison. An invertible generator intertwiner would preserve the spectrum, contradicting Eq. (29). No probability comparison between arbitrarily chosen c and a preparations is implied: without a physical intertwiner they do not specify one canonically transported experiment. Here the full matrix algebra would allow energy-gap-dependent evolution to be examined, but no claim about a restricted experimental algebra is inferred from spectrum alone.

The two patterns in the same synthetic carrier are

$$\begin{aligned}&\Pi_a^{(3)} \neq \Pi_b^{(3)},\ R_a \sim R_b \quad \Longrightarrow \quad \text{projector freedom can be representational,} \\ &\Pi_a^{(3)} \neq \Pi_c^{(3)},\ R_a \not\sim R_c \quad \Longrightarrow \quad \text{admissibility need not imply strong equivalence.}\end{aligned} \tag{30}$$

No claim is made that a particular motivating model realizes either pattern generically. The example isolates a structural distinction, not an ontology of multiple realizations.

5.3. Dimension Two: a Normalization Caution Only

A familiar PT-symmetric example [5,6]:

$$H_2 = \begin{pmatrix} i\gamma & 1 \\ 1 & -i\gamma \end{pmatrix}, \qquad G_{2,w} = \begin{pmatrix} 1 & w - i\gamma \\ w + i\gamma & 1 \end{pmatrix}, \quad |\gamma| < 1, \quad w^2 + \gamma^2 < 1. \tag{31}$$

For real γ, w, the metric eigenvalues are $1 \pm \sqrt{w^2 + \gamma^2} > 0$ and $H_2^\dagger G_{2,w} = G_{2,w} H_2$. The auxiliary norm need not be conserved: for initial auxiliary vector e_1, its derivative is 2γ at $t = 0$. The $G_{2,w}$-normalization is conserved. This family illustrates compatibility freedom for the generator; an additional fixed observable algebra may restrict w. The projective Gleason corollary does not apply in dimension two, and this example makes no Born-rule derivation or general no-signaling claim.

6. Scope and Non-Claims

Projection nonuniqueness concerns distinct admissible Π_α; metric nonuniqueness concerns distinct compatible G_α within a fixed sector. Neither specifies a probability distribution on projection labels. Sectorwise Born probability is conditional on a declared admissible sector and must not be

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interpreted as a measure over admissible projections. The equivalence proposition compares already admissible realizations and corresponding preparations. It proves neither that all sectors are equivalent nor that a single equivalence class exists. It assigns no realization probabilities, explains no realized-sector selection, and supplies no inter-sector transitions, observer or consciousness selection, or ontology of multiple sectors.

The normalization proposition uses a fixed sector and metric with a self-adjoint physical generator. It does not automatically extend to exceptional points, defective finite-dimensional generators, broken-PT regimes lacking a positive metric, or time-dependent projections and metrics. In particular a varying metric adds $\langle \psi \mid \dot{G}\psi \rangle$ to the derivative, so the fixed-metric identity alone would not establish conservation. Unbounded operators require the exact domain hypotheses stated above. A local current requires additional locality and kinetic-operator structure; global normalization does not prove a model-independent local continuity law. Nor does this single-sector construction establish multipartite composition, locality or no-signaling for a complete theory.

The results concern only the probability structures of supplied admissible realizations. They do not construct or complete a motivating physical model, establish empirical validation, or classify its geometric projections. Projective measure and pure-preparation hypotheses remain explicit inputs; the construction is neither an unrestricted Born-rule derivation nor a solution of the measurement problem.

7. Conclusions

Strong operational equivalence of complete admissible projected realizations preserves probabilities on equivalence classes of pointed preparation and measurement data. Kernel, algebra, commutant and spectral-measure invariants provide necessary obstruction tests. For a full matrix observable algebra, irreducible physical representations and a common abstract dynamical element provide a sufficient finite-dimensional criterion. The dimension-three examples display both distinct equivalent projections and an individually admissible but strongly inequivalent sector. Individual admissibility therefore does not establish equivalence. Classification of concrete projected realizations in motivating models remains open. Neither metric uniqueness nor a probability measure on projection labels is supplied.

Data Availability Statement: No empirical datasets were generated or analysed. The synthetic finite-dimensional calculations are reproducible with the verification scripts supplied alongside the manuscript.

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References

1. Chien-Chih Chen, "Projected Scalar Green Functions and Schur Reduction of PT-Symmetric Quaternionic Principal-Symbol Deformations," *Particles* **9** (2026), 93. doi:10.3390/particles9030093.
2. M. Born, "Zur Quantenmechanik der Stoßvorgänge," *Zeitschrift für Physik* **37**, 863–867 (1926). doi:10.1007/BF01397477.
3. A. M. Gleason, "Measures on the closed subspaces of a Hilbert space," *Journal of Mathematics and Mechanics* **6**, 885–893 (1957).

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4. C. M. Bender and S. Boettcher, "Real spectra in non-Hermitian Hamiltonians having $\mathcal{PT}$ symmetry," *Physical Review Letters* **80**, 5243–5246 (1998). doi:10.1103/PhysRevLett.80.5243.
5. Y.-C. Lee, M.-H. Hsieh, S. T. Flammia, and R.-K. Lee, "Local PT symmetry violates the no-signaling principle," *Physical Review Letters* **112**, 130404 (2014). doi:10.1103/PhysRevLett.112.130404.
6. M. Znojil, "Is PT-symmetric quantum theory false as a fundamental theory?" *Acta Polytechnica* **56**(3), 254–257 (2016). doi:10.14311/AP.2016.56.0254.
7. A. Mostafazadeh, "Pseudo-Hermiticity versus $\mathcal{PT}$ symmetry II: A complete characterization of non-Hermitian Hamiltonians with a real spectrum," *Journal of Mathematical Physics* **43**, 2814–2816 (2002). doi:10.1063/1.1461427.
8. A. Mostafazadeh, "Pseudo-Hermitian representation of quantum mechanics," *International Journal of Geometric Methods in Modern Physics* **7**, 1191–1306 (2010). doi:10.1142/S0219887810004816.
9. D. Finkelstein, J. M. Jauch, S. Schiminovich, and D. Speiser, "Foundations of quaternion quantum mechanics," *Journal of Mathematical Physics* **3**, 207–220 (1962).
10. S. L. Adler, *Quaternionic Quantum Mechanics and Quantum Fields* (Oxford University Press, New York, 1995).
11. F. G. Scholtz, H. B. Geyer, and F. J. W. Hahne, "Quasi-Hermitian operators in quantum mechanics and the variational principle," *Annals of Physics* **213**, 74–101 (1992). doi:10.1016/0003-4916(92)90284-S.
12. H. B. Geyer, W. D. Heiss, and F. G. Scholtz, "The physical interpretation of non-Hermitian Hamiltonians and other observables," *Canadian Journal of Physics* **86**, 1195–1201 (2008). doi:10.1139/P08-060.
13. V. Moretti and M. Oppio, "The correct formulation of Gleason's theorem in quaternionic Hilbert spaces," *Annales Henri Poincaré* **19**, 3321–3355 (2018). doi:10.1007/s00023-018-0729-8; arXiv:1803.06882 [math-ph].
14. G. J. Murphy, *C^*-Algebras and Operator Theory* (Academic Press, 1990).

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